6.8 Shapes, maps and graphs: activities

← Unit overview

1. Map of our town

You need: a paper map of your town or neighbourhood (from the town hall, tourist office, or printed from an online map), a pencil, a ruler.

1. If the map has no grid, draw one: letters along the top, numbers down the side.
2. Find your house. Which square is it in? The school? The supermarket? Grandma's house?
3. Use the scale bar: measure from home to school with a ruler and work out the real distance.
4. Check it: cycle or walk the route and compare with a bike computer or phone.

Talk about it: "Was the real route longer than the straight line on the map? Why?"

2. Bedroom floor plan

You need: squared paper, a tape measure, a pencil.

1. Measure the bedroom and the big furniture (bed, desk, wardrobe).
2. Draw a floor plan from above: 1 square = 50 cm (or 1 square = 1 m for a simpler version).
3. Mark the door and the window.
4. Design challenge: "Where else could the bed go? Does it fit?" Cut out paper furniture to move around.

Talk about it: "Why is a floor plan drawn from above, not from the side?"

3. Block buildings

You need: 10–20 blocks, Lego bricks or sugar cubes, squared paper.

1. Build a small building (5–8 blocks).
2. Draw it from the front, from the side, and from the top. For the top view, write in each square how many blocks are stacked there.
3. Swap: you draw a top view with numbers; he builds it. Then check the front view together.
4. Mystery: show only the front view and top view. Can he build it? Is there more than one way?

Talk about it: "Which view tells you the most? Which view hides things?"

4. Cube nets

You need: squared paper (2 cm squares are easiest), scissors, tape.

1. Unfold a small cardboard box (a cereal or tea box). Look at the net.
2. Draw a cross-shaped net of 6 squares, cut it out and fold it into a cube.
3. Draw other arrangements of 6 squares. Predict first: will it fold into a cube? Then cut and test.
4. Keep the ones that work on one sheet and the ones that don't on another.

Talk about it: "What's a clue that a net won't work?" (For example, four squares meeting at one corner, or five in a row.)

5. Our own graph

You need: squared paper, a pencil, a week.

Choose something to count every day for a week: the temperature at breakfast (from a thermometer or weather app), the number of bikes in front of the school, the minutes spent reading.

- Temperature or time-based data → draw a line graph.
- Counts → draw a pictogram (make up a symbol: one bike = 2 bikes).

Then ask questions about it, and try a "sneaky" graph: draw a bar chart where the axis starts at 50 instead of 0 and see how it changes the picture.

Talk about it: "How can a graph trick you?"

Conversation prompts (any time)

- "If you were a bird above our house, what would you see?"
- "Which way is north on this map? How do you know?"
- "Why do you think maps use a scale instead of real size?"
- "Where have you seen a graph recently? What was it trying to tell you?"